

# 5

## ISSUES IN THE TEACHING OF SPANISH LIQUID CONSONANTS<sup>1</sup>

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### 5.1 Introduction

The vast majority of Spanish varieties exhibit three **liquids** at the phonemic (see **phoneme**) level, namely, two **rhotics**, a **voiced alveolar tap**, /ɾ/, and a voiced alveolar trill, /r/, along with one voiced **lateral approximant**, /l/. Of particular interest for **L2** liquid production is that English only has one of these, the voiced lateral approximant, at the phonemic level. In recent years, a number of studies (e.g., Díaz-Campos 2004; Lord 2005; Waltmunson 2005; Face 2006; Colantoni and Steele 2006; Johnson 2008; Olsen 2012; Rose 2012; Scarpace 2014; Solon 2015, 2016) have treated the L2 production of Spanish liquids, primarily with regard to students whose **L1** is English. We must not only further our knowledge from the novel findings of these recent studies, but also consider their implications for the classroom. To this end, in Section 5.2, I first give a brief overview of the Spanish liquid inventory. Additionally, in Section 5.3, I discuss English liquids before reviewing the literature on L2 production of Spanish liquids. Then, in Section 5.4, I offer practical suggestions for the classroom based on previous findings and my own experience with teaching **phonetics** and **phonology**, before concluding the chapter in Section 5.5.

### 5.2 Spanish liquids

**Sonorants** are produced with periodic airflow that is neither interrupted nor turbulent. Sonorants are, by nature, typically voiced cross-linguistically and are comprised of **vowels**, **glides**, nasals (see **nasal-stop class**), laterals, and rhotics; the final two categories are often combined and referred to as *liquids*.

With direct regard to Spanish, linguistic studies have traditionally noted three liquids at the phonemic level, namely, one lateral and two rhotics, which are associated with

the **graphemes** <l>, <r>, and <rr>, respectively (Massone 1988, 15). In what follows, I elaborate on each one of the Spanish liquid phonemes in articulatory (see **articulatory phonetics**), acoustic (see **acoustic phonetics**), phonemic, and allophonic (see **allophone**) terms. Given that the focus here is on L2 teaching and production, extensive non-contrastive, phonetic detail falls mostly outside of the current chapter's focus, and as such, is not discussed at length.

## 5.2.1 Spanish lateral approximants

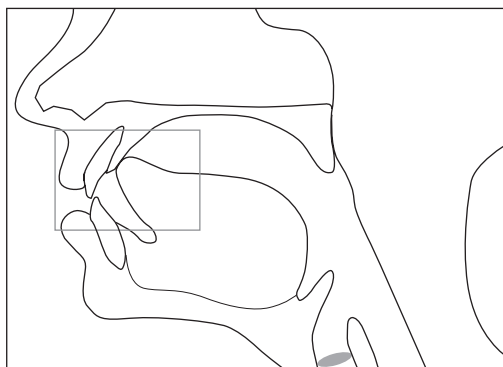
### 5.2.1.1 Articulatory properties of the lateral approximant phoneme

The Spanish voiced alveolar lateral approximant is produced when the vocal cords vibrate and the **tongue** tip strikes the alveolar ridge. As it maintains contact with the alveolar ridge, airflow passes without turbulence along one or both sides of the tongue, and the tongue dorsum does not raise toward the **velum**, as seen in Figure 5.1.

### 5.2.1.2 Acoustic properties of lateral approximants

Lateral approximants have similar **formant** structure and are characterized by faint formants in the lower frequencies. Average values for one variety, Peninsular Spanish, were 328 Hz for the first formant (F1), 1,561 Hz for the second formant (F2), and 2,587 Hz for the third formant (F3) (Quilis et al. 1979). Massone (1988, 22) observed an average of 400 Hz for F1, 1,550–2,000 Hz for F2, and around 2,700 Hz for F3.

In Figure 5.2, I show a **waveform** and **spectrogram** of *los*. In the figure, one can see that the initial **consonant** (i.e., the lateral approximant) and the following **vowel** are voiced.



**FIGURE 5.1** Articulatory production of a Spanish lateral approximant.

Source: The figures for discussion on articulatory properties are taken from UIRF (n.d.).

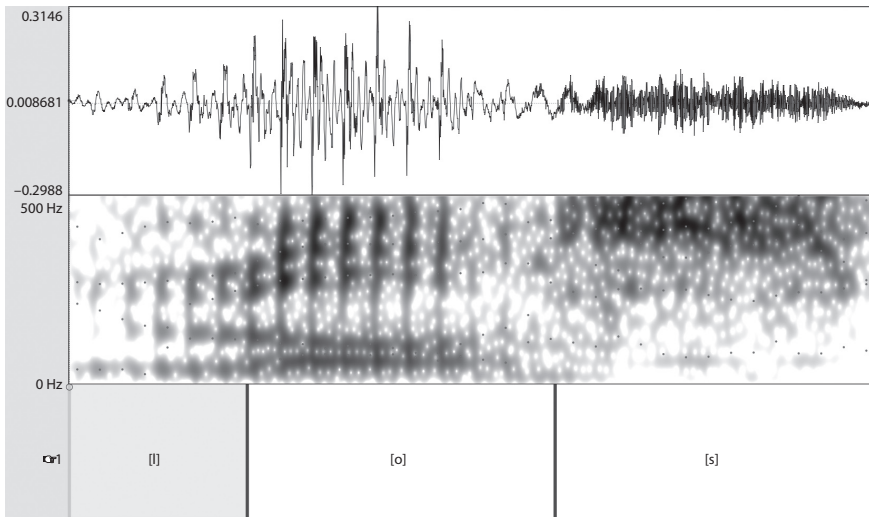


FIGURE 5.2 The waveform and spectrogram of a production of *los*.

### 5.2.1.3 Spanish lateral approximant phonemes

At the phonemic level, when it is not part of a consonant cluster, the Spanish voiced alveolar lateral approximant, /l/, is found in **onset** position for the word-initial (1a), and word-medial (and utterance-medial) intervocalic contexts (see **phonetic context**) (1b), it is also found in **coda** position when it is in word-final position before a pause (1c).

- |        |                |              |
|--------|----------------|--------------|
| (1) a. | <i>lámpara</i> | [ˈlam.pa.ra] |
| b.     | <i>alas</i>    | [ˈa.las]     |
| c.     | <i>animal</i>  | [a.ni.ˈmal]  |

In addition, /l/ occurs in consonant clusters. When it is the second member of a consonant cluster, it is in a (**tautosyllabic**) complex onset, as in the examples in (2).

- |        |               |            |
|--------|---------------|------------|
| (2) a. | <i>planta</i> | [ˈplaŋ.ta] |
| b.     | <i>blusa</i>  | [ˈblu.sa]  |
| c.     | <i>gloria</i> | [ˈglo.ɾja] |
| d.     | <i>claro</i>  | [ˈkla.ro]  |

The Spanish lateral approximant is also found in coda position, both word- and utterance-medially, before a following onset consonant (i.e., in a **heterosyllabic** consonant cluster), as displayed in (3).

- (3) a. *alma* [ˈa.l.ma]  
 b. *el niño* [el.ˈni.ɲo]  
 c. *el ratón* [el.ra.ˈɬon]

In addition to the voiced alveolar lateral approximant, some varieties include a voiced **palatal** lateral approximant, /ʎ/, which is phonemically distinct from a voiced palatal **fricative**, and diachronically, these two phonemes were in opposition (see **contrast**), as shown in (4).

- (4) a. *haya* [ˈa.ja]  
 b. *halla* [ˈa.ʎa]

This distinction is found in Old Spanish, Castilian Spanish, the Andean Region, and in parts of Paraguay.

#### 5.2.1.4 Lateral approximant allophony<sup>2</sup>

Spanish evidences a series of allophones due to **assimilation** in **place of articulation**, where a lateral approximant takes the same place of articulation as the following **coronal** consonant; this process applies obligatorily within words but tends to be conditioned by speech style and rate across word boundaries. Examples of the assimilatory process are illustrated in (5).

- (5) a. *alto* [ˈaɫ̺.ɬo]  
 b. *colchón* [koɫ̺.ˈtʃon].  
 c. *el yeso* [eʎ̺.ˈje.so]  
 d. *alzar* [aɫ̺.ˈθar] (in Castilian Spanish)

As shown above, a lateral approximant before the **voiceless dental** consonant in *alto* becomes a voiced dental lateral approximant in (5a), a lateral approximant before an alveo-palatal consonant produces a voiced alveo-palatal lateral approximant in (5b), and before a palatal consonant, a palatal lateral approximant in (5c). Finally, in Castilian Spanish, a lateral approximant before an **interdental** consonant is realized as an interdental lateral approximant in (5d).

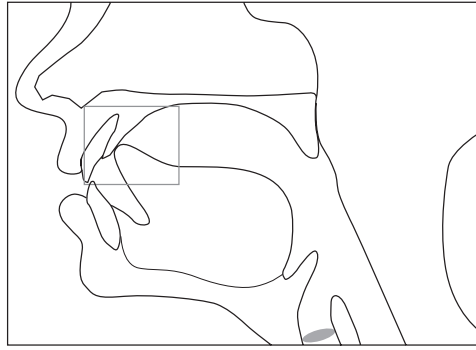
### 5.2.2 Spanish rhotics

#### 5.2.2.1 Articulatory properties of rhotics

Spanish presents two rhotic phonemes, the voiced alveolar tap, /ɾ/, and the voiced alveolar trill, /r/. The tap is characterized by a rapid upward strike of the tongue tip against the alveolar ridge, the tongue tip remains in contact with the alveolar ridge for 18–25 milliseconds (ms). Figure 5.3 depicts this articulatory motion.

The Spanish trill, it should be noted, is not a simple repetition of tap gestures, but rather a different articulatory mechanism altogether. The tongue must be held in a



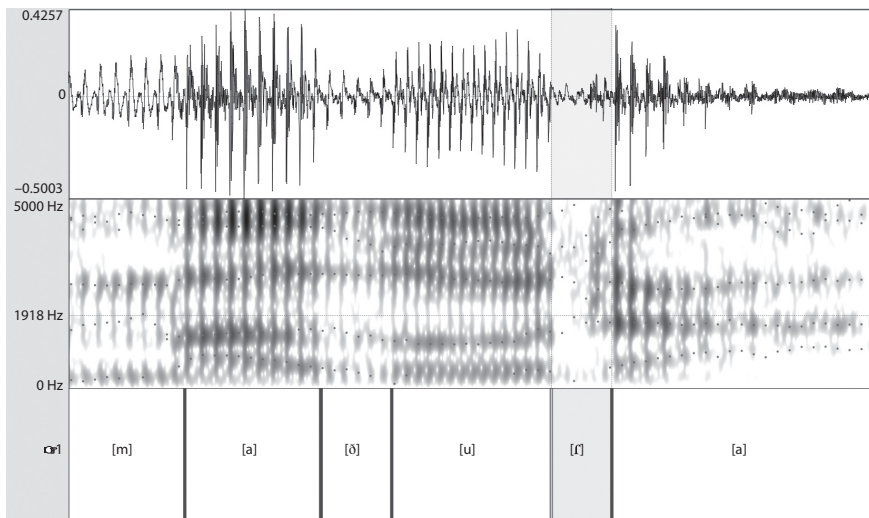


**FIGURE 5.3** Articulatory production of a Spanish tap.

precise position to create the aerodynamic conditions for passive tongue tip vibration by the Bernoulli effect; speakers are not intentionally flicking their tongue tip against the alveolar ridge each time, but rather, the lingual vibration is passive and induced under a sufficient airstream (Recasens 1991, Recasens and Pallarés 1999, Martínez Celdrán and Fernández Planas 2007, 147–151, as cited in Bradley in press).

#### 5.2.2.2 Acoustic properties of rhotics

In terms of its acoustic properties, the tap is quite similar to a **stop consonant**, though it often lacks a burst typically found after stops. Massone (1988, 23) observed an average of 500 Hz for F1, 1,200–1,600 Hz for F2 (depending on the following vowel), and 2,200 Hz for F3. In Figure 5.4, we see an intervocalic tap in *madura*.



**FIGURE 5.4** A canonical tap in *madura*.

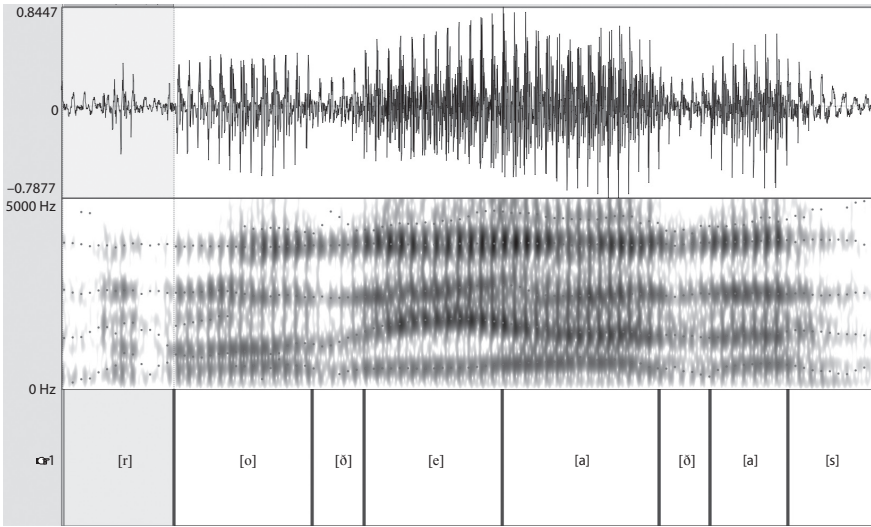


FIGURE 5.5 A production of [r] in *rodeadas*.

In the case of the trill, there are two (or more) **occlusions**, between which is a vowel-like fragment, as seen in Figure 5.5 for the word *rodeadas*. In the figure, one can see an occlusion, followed by the vowel-like fragment, and a second occlusion. In this case, a trill is obligatory for the grapheme <r> in word-initial position (for further discussion, see Section 5.2.2.3).

Finally, I note that Spanish evidences a **svarabhakti vowel** (i.e., intrusive vowel) between a tap and its adjacent consonant; it may also occur in word-final position (Schmeiser 2006). Svarabhakti vowels are unique in two ways: first, they do not occupy the position of nuclear vowel (see **nucleus**), second, they often have similar formant structure as the nuclear vowel in the same **syllable** (Quilis 1970). In (6), we see a list of examples for all three contexts, namely, a tautosyllabic consonant cluster in (6a), a heterosyllabic consonant cluster in (6b), and utterance-final position in (6c).

- (6) a. *pron.to* [p<sup>ə</sup>r]  
b. *par.te* [r<sup>ə</sup>t]  
c. *Voy a escribir* [r<sup>ə</sup>]

### 5.2.2.3 Spanish rhotic phonemes

In Standard Spanish, there are two rhotics at the phonemic level. The voiced alveolar tap, /r/, generally exhibits a tap allophone in intervocalic position for word-medial (and utterance-medial) contexts, and in word-final position before a pause, as seen in (7).

- (7) a. *cara* [ˈka.ra]  
 b. *hablar* [a.ˈβlar]

It also occurs in consonant clusters, most commonly, as the second element in a tautosyllabic consonant cluster, either in word-initial or word-medial position (8a–b), it also occurs as the first element in a heterosyllabic consonant cluster (8c).

- (8) a. *treinta* [ˈtɾei̯n̩.ta]  
 b. *atrás* [a.ˈtɾas]  
 c. *carta* [ˈkar.ta]

In addition, it may occur as the first element in a tautosyllabic consonant cluster (and the second element is always an /s/), as seen in (9).

- (9) *perspectiva* [pers.pek.ˈt̪i.βa]

With regard to the voiced alveolar trill, /r/, it is prominently produced in three contexts (10a, 10b, and 11) and is possible in (extremely) emphatic speech in three others (12a, 12b, and 13). It is produced in word-initial and intervocalic positions (in the latter, it is in phonemic contrast with the tap), as seen in the following examples in (10).

- (10) a. *rosa* [ˈro.sa]  
 b. *carro* [ˈka.ro]

In addition, a trill is produced in consonant clusters in which the rhotic follows a **homorganic** consonant (i.e., /n, l, s/), as in (11).

- (11) a. *Enrique* [en.ˈri.ke]  
 b. *Israel* [iz.ˈra.el]  
 c. *alrededor* [al.re.ðe.ˈðor]

In Spanish, a trill may also be produced in utterance-final or word-medial positions before a consonant (i.e., in those environments in which it is not in phonemic contrast with the tap) in emphatic speech, as in the following examples in (12).

- (12) a. *hablar* [a.ˈβlar]  
 b. *harpa* [ˈar.pa]

Finally, I note that in extremely emphatic speech, a tap can be realized as a trill when it is the second member of a tautosyllabic consonant cluster, as in (13).

- (13) *prensa* [ˈpren.sa]

### 5.2.2.4 Rhotic allophony

To discuss rhotic allophony in great detail would easily merit (at least) an entire article; however, due to space constraints, I quickly mention a few common variations. In the case of the tautosyllabic consonant cluster, a non-tap allophone can arise in some varieties, especially following the voiceless dental stop, after which the tap is realized as a voiceless alveolar approximant (14a); this approximant appears most commonly in utterance-final position (14b). A non-tap allophone also occurs when the tap is the first member of a consonant cluster (i.e., the heterosyllabic environment), though this allophone is not nearly as common, as it occurs principally in Andean Spanish (14c). Finally, with regard to the trill, a voiceless approximant allophone is common in many varieties (14d).

- |      |    |                       |                   |
|------|----|-----------------------|-------------------|
| (14) | a. | <i>otro</i>           | [ˈo.t̪o]          |
|      | b. | <i>vamos a hablar</i> | [ˈba.mo.sa.ˈβla̠] |
|      | c. | <i>cuervo</i>         | [ˈkue̠.βo]        |
|      | d. | <i>perro</i>          | [ˈpe.̠o]          |

## 5.3 Spanish liquid production by second language learners<sup>3</sup>

Spanish Linguistics has a long history of study dedicated to liquid production. Early studies (e.g., Navarro Tomás 1918; Gili Gaya 1921; Alonso 1925) concentrated on articulatory and acoustic traits, while later studies (e.g., Quilis 1970, 1993; Massone 1988) furthered the discussion by offering empirical and spectrographic analyses. Since the turn of the century, a number of studies (e.g., Blecua Falgueras 2001; Bradley and Schmeiser 2003; Bradley 2004, 2005; Lewis 2004; Colantoni and Steele 2006; Proctor 2009, 2011) have analyzed liquid production in more theoretical terms.

With specific regard to second language acquisition (SLA), Sacks's (1962) seminal study included liquid productions of 29 participants at the Summer Language Institute at the University of Wisconsin-Madison. Participants were asked to make recordings during the first and final weeks of the program. In this study, he referred to the liquids as the *American l* and *American r* (1962, 292) and discussed why students might not produce the correct form. For the lateral approximant, he noted that **velar coarticulation** (which produces a **dark l**) was difficult to stop in word-final position. In addition, for the tap, which takes the longest to acquire (in his estimation), students tended to produce a **retroflex**, a voiced alveolar stop or a trill.

In recent years, researchers have shown a heightened interest in liquid production as it pertains to SLA. In what follows, I discuss these studies in terms of each liquid. Due to space limitations, I only discuss studies that specifically deal with Spanish liquid production of students whose L1 is English. I concentrate on learners whose L1 is English primarily because the majority of previous research focuses on these learners. The reader will benefit from what follows on English liquids before we treat acquisitional studies in Section 5.3.2.

### 5.3.1 English liquids

In order to understand L2 liquid production, we must not only teach canonical Spanish liquid pronunciation, but also have a general understanding of L1 liquid production to ascertain to what degree there is transfer. As such, the aim in this section is to provide a very basic overview of English liquid production to have a base from which we can discuss and fully understand the literature review in the remainder of this section, which provides a point of departure for the practical suggestions in Section 5.4.

#### 5.3.1.1 Laterals

American English has a lateral approximant, but it differs from Spanish in two ways: first, in terms of phonetic properties, American English has a “darker” lateral approximant (in all contexts) due to two co-occurring gestures, namely, a consonantal tongue tip gesture and a vocalic tongue-dorsum gesture (Sproat and Fujimura 1993, as cited by Barlow 2014, 2)<sup>4</sup>, second, American English exhibits allophonic distribution (see **complementary distribution**) in a **velarization** rule in which [ɫ] is produced in the syllable **rhyme** and [l] is produced in the syllable onset (Barlow 2014, 2), as seen in (15).<sup>5</sup>

- |      |                |        |
|------|----------------|--------|
| (15) | a. <i>meal</i> | [miɫ]  |
|      | b. <i>lamp</i> | [læmp] |

I emphasize that, though (15a) shows the velarized lateral, the clear lateral (see **clear l**) in (15b) is still darker than its Spanish counterpart.

#### 5.3.1.2 Rhotics

In terms of rhotic production, American English exhibits one, the voiced approximant, /ɹ/, in all environments, there is no trill. In (16), I offer word-initial (16a), word-medial (16b), and word-final (16c) examples, respectively, while also including an example in a complex onset in which there is progressive devoicing after a voiceless stop (16d).

- |      |                |         |
|------|----------------|---------|
| (16) | a. <i>red</i>  | [ɹ]ed   |
|      | b. <i>Kara</i> | Ka[ɹ]a  |
|      | c. <i>car</i>  | ca[ɹ]   |
|      | d. <i>tree</i> | t[ɹ̥]ee |

American English does have a tap,<sup>6</sup> but it functions as an allophone of /t/ and /d/ production and is never associated with the grapheme <r>. As Turk (1992, 104) points out, there are three rules conditioning the distribution of the tap in American English: 1. after a stressed vowel and before an unstressed one (17a), 2. between

two unstressed syllables (17b), and 3. word-finally between two vowels (17c, vowels can be stressed or unstressed). It should be noted that only the first rule is obligatory.

- |      |                       |               |
|------|-----------------------|---------------|
| (17) | a. <i>medal</i>       | me[r]al       |
|      | b. <i>provacative</i> | provaca[r]ive |
|      | c. <i>not at all</i>  | no[r]a[r]all  |

Though the emphasis of this chapter for L1 production is American English, it is important to note that rhotic variation in students' L1 will affect how they produce Spanish rhotics. Many varieties of British English include a rhotacized vowel instead of an approximant, as in *car* [kɑː], and thus, students from these varieties may tend to produce this segment in Spanish as well. Furthermore, in some varieties in the United Kingdom, a tap is used intervocalically (e.g., *Terry* ['tɛ.ri]). Finally, some speakers of Scottish English may employ a tap in tautosyllabic consonant clusters (e.g., *crush* [krʌʃ]). Teachers should keep such variation in mind when addressing rhotics in their L2 Spanish courses.

### 5.3.2 Previous research on the acquisition of Spanish liquids

Before reviewing the following studies, I stress that while the problematic aspects that are highlighted in L2 studies deal with L1 English speaking learners, they motivate teaching suggestions (in Section 5.4) that can be implemented with speakers of any L1 that has a different liquid inventory and distribution than Spanish.

#### 5.3.2.1 Spanish lateral approximant

The past two decades of research have expanded our knowledge of how native English-speaking learners acquire laterals in L2 Spanish. Research has predictably focused on the two main areas in which the two languages differ in terms of the lateral approximant, namely: 1. a darker lateral approximant, and 2. velarization. Both of these features are found in American English and not in Spanish.

In Díaz-Campos's (2004) study of L2 consonant production, he quantitatively analyzed four areas commonly discussed in textbooks on Spanish pronunciation, namely, the aspiration of voiceless stops (see **aspirated stop**), voiced intervocalic fricatives, the velarization of word-final [l], and the palatal nasal. There were 20 participants in one group (AH = at home) and 26 participants in another group (SA = study abroad) that completed a ten-week program in Alicante, Spain. Subjects were asked to read a passage aloud twice, once at the beginning and once at the end of the program. Then, 368 tokens were analyzed for the word-final [l] environment, with the majority of participants exhibiting the non-native (i.e., velarized) variant; however, both groups were able to improve in the production of word-final [l] over time.

Solon's (2015) work considered Spanish native speakers' foreign accent rating of word list production data from 20 English speakers and four native speakers, with

particular attention on the dark, or velarized, lateral production found in English. She considered the role of F2 values, which correlate to velarization, and segment duration in foreign accent perception. There were 38 subjects, all from Mexico and native speakers of Spanish, who were asked to complete a foreign accent-rating task. Word-medial tokens with coda /l/s with low F2 values (i.e., velarized lateral production) were rated as carrying significantly more foreign accent than high F2 tokens. In addition, Solon also found that, for both word-medial and word-final positions, /l/s with longer duration were rated as significantly more foreign sounding.

Solon's (2016) study on lateral production of L2 learners of Spanish was comprised of 85 subjects whose native language is English. Data suggest that higher-level learners are able to shift from the bimodal distribution in their L1 (i.e., a clear lateral approximant in onset position and a dark lateral approximant in coda position) to the monomodal distribution of Spanish.

In sum, studies on lateral production evidence (at least) one prominent finding, namely, that students are able to make linguistic gains toward L2 production over time, that is, as they become more proficient in Spanish, velarization decreases.

### 5.3.2.2 *Spanish rhotics*

Since Major's (1986) study, there has been a steady stream of publications treating Spanish rhotic production by L1 speakers of English. Given that many of these publications include both the tap and the trill, in what follows, I discuss various recent studies in chronological order and do not separate them by rhotic type.

Major's (1986) empirical study on Spanish rhotics with four participants in a short, intensive period found that students increased their accuracy of both Spanish rhotics as they progressed in the language (i.e., gained more exposure). That said, his study did show that participants improved more with regard to the Spanish tap than the trill.

Elliott's (1997) study focused on how phonological instruction related to improved pronunciation. In addition, it also considered if formal instruction was more beneficial for word repetition, sentence repetition, word reading, or spontaneous production. Experimental group subjects received 10–15 minutes of formal instruction and practice of Spanish pronunciation for a total of 21 class periods, and took a pre-test (in the second week) and a post-test (at the end of the semester). The results indicated that the experimental subjects showed significant improvement of the two rhotics. With regard to the tap, experimental subjects learned to pronounce the intervocalic <r> by replicating the flap in words like *butter*, *city*, and *water*, which the author explained as a case of positive transfer. For the trill, the improvement was attributed to students learning a new sound not present in the L1.

Reeder's (1998) study treated trill production in beginning, intermediate, advanced, and very advanced groups of L2 learners. In terms of data collection, he used a questionnaire, a perception test, and recordings of the subjects' production.

The study showed clearly defined progress in all four groups in terms of number of taps and frequency of trill production.

Lord (2005) looked at the effects of explicit instruction and self-analysis on L2 pronunciation, including the trill. The subjects were 17 undergraduates enrolled in an upper-division, Spanish phonetics class. The gains made in trill production when comparing the pre-instruction and post-instruction conditions were statistically significant.

Waltmunson's (2005) dissertation analyzed the relative degree of difficulty of L2 Spanish /d, t, r, r/. He found that /t/ was the least difficult, whereas the trill was the most difficult. Crucially, he found that word position and vowel height had a strong effect on scores for the trill. In terms of word position, word-medial trill production was higher (in every group) than word-initial production. In light of vowel height, trill production was lowest in the high vowel environment, higher in the mid vowel environment, and highest in the low vowel environment. For the tap in complex onsets, there was a strong effect of syllable position on accuracy scores.

Face's (2006) in-depth empirical study on Spanish rhotic production in intervocalic context again found that participants improved. His study evidenced a much higher success rate for taps than for trills. Crucially, his study also found that, though there was an increase in accuracy for both rhotics, the development process was different for each rhotic. More specifically, for the tap, there was a higher level of accuracy, and most non-target productions were due to L1 transfer; however, for the trills, there were fewer cases of L1 transfer, but there was an increase in over-generalization of the L2 tap.

Colantoni and Steele (2006) focused on L2 production of Spanish stop + liquid clusters. They found that only one of the ten participants mastered the stop + liquid cluster. The authors questioned previous claims regarding maturation of post-puberty learners, given that one participant did, in fact, master the production in question. They then posited that learners' difficulty in parsing variability may inhibit accurate learning in that the more categorical a phonetic property is, the more likely it will be mastered, at least by some learners.

Johnson's (2008) dissertation examined L2 production of the Spanish trill in four student groups: first semester, fourth semester, intermediate, and those who held at least a BA in Spanish. He conducted two experiments: the first was to determine the frequency of trilling in L2 Spanish learners and the second was an aerodynamic study of native and non-native trilling. For the first experiment, he looked at trill, tap, and American /ɹ/ for trill production. His general overall results reported that trill production was low at first, when the retroflex rhotic was common, but as students progressed to higher levels, retroflex production decreased as tap and trill production increased. In the second experiment, Johnson also found that non-native speakers were not the same as native speakers in aerodynamic terms.

Olsen (2012) conducted a detailed empirical study on the L2 acquisition of Spanish rhotic production by L1 English speakers. His results showed a significantly high percentage of accurately produced taps in Spanish when they appeared in the same articulatory context (i.e. following a stressed syllable and in the onset of an



unstressed syllable) that produces taps in English (2012, 62–70). Olsen found that, for taps, the most important factor was experience; however, English rhotic articulation did influence the accuracy of tap production. In terms of trill production, Olsen's data suggest that speakers who use a retroflex-like articulation in English are more likely to accurately produce the Spanish trill than speakers who use a bunched-like (i.e., contract their tongue tip back and raise the tongue dorsum toward the palate) articulation (2012, 68). That said, the success rate for trill production was quite low.

Rose (2012) referred to her previous study (2010), in which she observed that American-English speakers can discriminate between the Spanish /r/-/r/ and /r/-/t/ contrasts significantly better than the /r/-/d/ contrast. In the study, the subjects had little or no Spanish experience and completed a cross-language mapping task; this type of task aimed to ascertain how sounds were determined in the L2. Her findings suggest that this contrast is particularly difficult because students categorize the Spanish /r/ and /d/ to similar phonemic categories in English. She also suggests that the Spanish /r/ is the only consonant that is assimilated to a single category in English.

Scarpace's (2014) perception study included both tap and trill production in terms of lexical encoding. He hypothesized that the word-internal contrast should be the easiest to detect, and that word-boundary conditions should be more difficult given the orthographic transparency (see **shallow orthography**) present in the former but not the latter position. There were two groups, a control group of native speakers and an experimental group. For the experimental group, there was an increase in accuracy as proficiency level increased, and contrasts at the word boundary were more difficult to acquire.

In sum, with regard to Spanish rhotic production, recent studies suggest that learners obtain a higher percentage of target production for the tap and trill as their proficiency increases. In the case of the tap, this is due in part to overt instruction and positive transfer. In the case of the trill, many learners obtain a higher percentage of target production in stages, from the retroflex [ɻ] to the tap, and finally, to the trill. In addition, for many, the intervocalic trill (especially word-medial position) is easier to produce, followed by word-initial position. Finally, surrounding vowel height affects trill production in that trill production increases from high to mid and mid to low vowels.

## 5.4 Practical applications in the classroom

In this section, I offer practical suggestions that teachers can use in the classroom. I emphasize that, though the suggestions are geared toward L1 English learners of Spanish, they can indeed be applied to L1 speakers of any language that are learning Spanish. I have tried to offer suggestions that are flexible and would adapt well to Spanish classrooms in which the L1 is not English.

I would also like to emphasize that, alongside these suggestions, teachers should consider incorporating technology (see Chapter 10); one way I do this in my classes

is by using a set of recordings in which speakers from different countries read a passage. In my experience, students appreciate that the passage is always the same, but like that we are able to emphasize different points, depending on the lesson.

### 5.4.1 Spanish lateral production

#### 5.4.1.1 Lateral approximant

In general, we know that English exhibits allophonic distribution in that there is a clear *l* in onset position and a dark *l* in coda position, we also know that Spanish does not have this process. It is pertinent, then, that learners are: 1. shown the difference between the two languages, and 2. practice meeting the target for lateral production without raising the tongue dorsum toward the velum in both onset and coda positions. I recommend starting with a word list in which the lateral is in onset position, as is the case in the sample words in (18). By starting in this manner, learners will gain confidence as they successfully meet their target production.

|      |                |               |
|------|----------------|---------------|
| (18) | <i>lámpara</i> | <i>planta</i> |
|      | <i>largo</i>   | <i>pleno</i>  |
|      | <i>lágrima</i> | <i>flor</i>   |
|      | <i>luz</i>     | <i>flauta</i> |
|      | <i>loro</i>    | <i>gloria</i> |

The next step is to practice /*l*/ target coda production, taking care *not* to raise the tongue dorsum toward the velum. Before practicing with this list, I find it particularly helpful to show students an example of what a beginner would sound like if velarization occurred, and then show them how to transition to the target production; for example, when students see that *tal* in Spanish and *tall* in English are often pronounced the same way by beginners, they are then curious to know how to use the more native-like construction. In this particular comparison, both *tal* and *tall* would be [tʰaɫ]. By showing learners how to refrain from raising their tongue dorsum toward the velum (and removing the burst after the stop, see Chapter 2), they can learn to produce [t̟aɫ] while also applying this articulation to other words with /*l*/ in the same context. At this point, they can practice another word list, such as the one in (19), but this time with the /*l*/ in coda position.

|      |            |                 |
|------|------------|-----------------|
| (19) | <i>tal</i> | <i>talco</i>    |
|      | <i>sol</i> | <i>soldado</i>  |
|      | <i>cal</i> | <i>caldo</i>    |
|      | <i>mal</i> | <i>maldad</i>   |
|      | <i>Gil</i> | <i>jilguero</i> |

For more intermediate and advanced students, teachers should teach the place of articulation assimilation rule for lateral approximants. This is particularly relevant

for /t/ and /d/ because they are always dental in Spanish and alveolar in English. For words like *alto* and *caldo*, students need to practice producing both the lateral approximant and the dental stop with dental rather than alveolar contact. Then, students can move on to the alveo-palatal region with words like *colchón*, and finally, to the palatal region (which I elaborate on below). By showing students processes such as the place of articulation assimilation rule for the lateral approximant, they become more proficient in their pronunciation and gain confidence at the same time. A word list for the above environments could include words such as those in (20).

- |      |                |                 |
|------|----------------|-----------------|
| (20) | <i>alto</i>    | <i>al tren</i>  |
|      | <i>caldo</i>   | <i>soldar</i>   |
|      | <i>colchón</i> | <i>al chico</i> |

#### 5.4.1.2 Palatal lateral

Though previous research has concentrated on the lateral approximant, I would like to briefly discuss the palatal lateral, [ʎ]. In Old Spanish, this sound was a phoneme for the grapheme <ll>; however, today, for most varieties of Spanish, the expected production is the voiced palatal stop, [j], in utterance-initial position (and after coda laterals) and the voiced palatal fricative, [j], in word-medial, intervocalic context. Therefore, first and foremost, teachers in beginning courses should be quick to eliminate the tendency students initially have to produce a lateral approximant in words such as *llama*. They will need to quickly associate the graphemes <ll> and <y> as having the same phoneme.

Second, students need to understand that they do not have the voiced palatal stop in their L1, but it is quite close to the voiced alveo-palatal **affricate**, [dʒ]. A helpful exercise to begin pronunciation practice of this sound is to have students pronounce *juice* or *judge* so that they see that their upper and lower teeth touch to produce the word-initial sound. In the case of the stop, the tongue carries a heavier load with increased palatal contact and the upper and lower teeth indeed do not touch in this production.

In the case of the word-medial, intervocalic context, learners should focus on the production of the voiced palatal fricative, [j], in English; for example, though it is true that a semivowel, [j], is used to produce *yes*, we do employ a fricative production for words like *yeast* and *yield*.<sup>7</sup> Teachers should remind students that for words like *ella* and *orgullo*, the palatal fricative production should be used in lieu of a semivowel (or glide).

Finally, with specific regard to the palatal lateral in Modern Spanish, it occurs when producing a lateral before a palatal consonant, as in *el llanto*. Teachers should show learners that they already have a similar sound in English and can realize this by saying the word *million* a few times, followed by *el llanto*, while focusing on the [ʎ] sound.

### 5.4.2 Spanish rhotic production

#### 5.4.2.1 Tap

Recall that a Spanish tap is produced by a quick strike by the tongue tip against the alveolar ridge. The first thing that the teacher should do is to have students realize that they already produce this sound in their L1. Considering the Elliott (1997) study above, students should begin by pronouncing words such as *city* and *butter* so that they get a feel for the production, which will aid in positive transfer of the tap. Then, students can alternate languages when practicing with a word list; for example, one exercise I like to do is practice with the words in (21).

|      |         |             |
|------|---------|-------------|
| (21) | English | Spanish     |
|      | cuter   | <i>cara</i> |
|      | patter  | <i>para</i> |
|      | cedar   | <i>cera</i> |

I like alternating between the two languages (I try to have the words start with the same letter, but it is not necessary to do so) because students focus more on the production of each word, but a teacher could always have students say the English words first and then the Spanish words. What is important here is that learners: 1. realize that they already produce the sound, and 2. focus on positively transferring this sound that they already have, but for a grapheme other than <r>.

Another helpful example is Morgan's (2014) work, which contains a helpful two-part PowerPoint Presentation with an accompanying lesson plan. One effective way he designs the lesson is to emphasize that we already have the tap, the teacher begins by saying, "tadah!" but the student sees only that word on the slide in Spanish, *¡tará!* Another example that I particularly like is a slide with the English expression *caught a cold*, followed by a slide with just a picture of a snail, which in Spanish is *caracol*, and almost identical phonetically to the English expression. Thoughtful, practical presentations such as Morgan's can greatly improve students' awareness of L1–L2 differences, this is especially important given the impact orthography has on the learning process (see Chapter 11 of this volume). In the above case of *¡tará!*, students already have the sound in their L1, but it is an **unfaithful allophone** for two non-rhotic phonemes. In addition, as Rose (2012) points out, students have a more difficult time seeing the difference between a tap and a /d/ because the Spanish stop has an allophone [ð] with a similar variant in English that is actually a phoneme associated with the grapheme <th>.

Concerning rhotic production, we now know that some students will begin to use the tap successfully with more exposure. That said, by using teaching devices such as Morgan's (2014), students will be able to better associate the same sound with different graphemes between the two languages, especially in intervocalic position. For both tauto- and heterosyllabic consonant clusters, a trait of more

advanced learners is the native-like production of svarabhakti vowels. Teachers should explain to students that a tap is really a three-part articulation: phonation with the tongue in its resting position, followed by the characteristic quick strike against the alveolar ridge, followed by the tongue descending to its resting position as phonation continues. By offering this explanation, it may help those students who feel they first need to understand how tap production is defined to help them later achieve more native-like pronunciation.

#### 5.4.2.2 Trill

Finally, I discuss the rhotic that has no equivalent for L1 English learners (and is not present in many languages in general), the trill. In terms of orthography, in the case of rhotics, <r> often signals a tap, and <rr> often signals a trill, but this is not always the case. Students should clearly know the context in which a trill is produced before practicing. They know it is produced when a word contains <rr>; however, word-initial <r> is often not seen as a context in which a trill is produced. As such, it is not surprising that Scarpace (2014) found that subjects used the trill at low rates in word-initial position.

Once learners understand all of the environments for trill production (see Section 5.2.2.3), teachers should use Teschner's (1999) practice exercise, which is outlined in (22).

(22) Exercise to practice the Spanish trill:

- a. Pronounce the *single* vibrant several times without voicing it, that is, the result should be a voiceless tap.
- b. Maintaining the tongue exactly where it should be held for the articulation of the voiceless tap, *make all muscles tense*, including the stomach, chest and throat.
- c. Now *exhale* strongly a long current of air. The air current will make the tongue vibrate several times in the position in which it is being held.
- d. Now you can *voice* the articulation, which has been converted into a multiple vibration, created by following the three previous stages. The result is the alveolar multiple vibrant sound.

*(adapted from Teschner 1999, 160–162, translation by and cited in Johnson 2008, 218)*

Teachers should then have students actually practice the sound in words. Tongue-twisters are particularly good for trill practice and the one in (23) is my favorite.

- (23) *Erre con erre, guitarra,  
erre con erre, carril:  
rápido ruedan los carros,  
rápido el ferrocarril.*

We know from previous research that as students become more proficient, they will likely abandon L1 interference and are likely to move to a tap, and from there, some move on to the trill. Our job as teachers is to help them get from the tap to the trill, or if they cannot, perhaps teach them trill allophones evidenced in lectal variation. By ensuring that students know *when* it is produced in Spanish, showing them *how* to produce it, discussing challenging areas like vowel height and word position, and then having them practice many tongue-twisters like the one listed above, students will gain confidence and begin to produce trills more consistently.

Finally, I would like to note that in many varieties of Spanish, the trill is used much less. For those students who truly have a difficult time with trill production, teachers might consider, in a more advanced class, going over an approximant production that many native speakers use; for example, in the case of *perro*, some native speakers use an assibilated (see **sibilant**) rhotic, often voiceless, as in [ˈpe.ʝo].

## 5.5 Conclusion

In short, I began this chapter with a basic overview of liquid production in Spanish in acoustic, articulatory, phonemic, and allophonic terms. In addition, I later offered a brief summary of liquids in English. Once defined, I reviewed the literature that has treated teaching liquid production in the classroom. Finally, I gave tips for the classroom to improve liquid production. As shown in this chapter, one of the keys to improved production is to increase learner awareness, include plenty of articulatory practice, and make sure they know where sounds should be produced.

In conclusion, I have found that one of the best ways to improve liquid production is to have students record themselves and analyze their own Spanish (e.g., using word lists at the end of all chapters after Chapter 9 in Guitart 2004, 102–272). When they listen to native speaker production followed by their own, they increase their awareness that their production is non-native, and naturally are curious to know how they can become more native-like. Keep in mind, however, that Spanish liquids, especially the trill, are among the most difficult for students to attain native-like pronunciation. Though we must remain realistic in terms of our expectations of our students, it is my sincere hope that the tools and exercises presented in this chapter can help them improve and advance toward native-like liquid production.

## Notes

- 1 I am deeply indebted to the reviewers for their invaluable feedback on previous versions of this chapter, they have offered insightful comments and constructive criticism. In addition, I would like to thank Travis Bradley for his feedback on earlier drafts of this manuscript. All errors and shortcomings are indeed my own. This chapter is dedicated to Dr. George Zucker.
- 2 This section concentrates on allophony of the lateral approximant and, for many reasons, does not treat at length delateralization. That is, for most varieties of Spanish, the grapheme <ll> requires a palatal approximant or affricate, which is commonly referred

- to as yeísmo. For an in-depth discussion on different non-lateral productions of this grapheme, see Guitart (2004).
- 3 I have included studies that tested a Spanish liquid exclusively as well as studies that included a liquid.
  - 4 Proctor's ultrasound-based work on Spanish (and Russian) liquids shows that both gestures are present universally in the class of liquid consonants. In the case of Spanish laterals, he notes that the advanced dorsum is a result of an intrinsic lingual target (2011, 462).
  - 5 Given that the lateral is clear in onset position, it would also be clear when it is the second element in a tautosyllabic consonant cluster (e.g., "play") and in word-medial intervocalic position (e.g., "caller").
  - 6 It should be noted that some linguists differentiate between a flap and a tap, for them, the articulatory production is different for a flap (in the English word, "tadah!") and the tap (in the Spanish word, ¡tará!). Ladefoged states, "A flap is therefore distinguished from a tap by having one articulator strike against another in passing while on its way back to its resting position, as opposed to striking immediately after leaving its resting position" (1971, 51). Whether you refer to them both as a tap or you differentiate between them (though in current IPA usage, the symbol [ɾ] is used for both), this exercise will be of value.
  - 7 I employ the term *semivowel* here to denote a sound that is phonetically like a vowel, but whose place in the syllable structure is that of a consonant.

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